

Session 6: The capstone — writing and defending a one-page AI (artificial intelligence) project brief

Saturday 26 September 2026, 11:00 AM to 12:30 PM EDT (Eastern Daylight Time). Videos V58 to V60. Practice session, student edition.

Student edition: the questions are here; the answers are discussed in the session.

Questions: Because of the interpretation delay, please send your questions in the Zoom chat, or better, after the session on the WhatsApp number you already have.

The interactive exercises live in the professor's edition and are shown on screen during the session. Each one is described here, with every control, its range and what it stands for.

Block 1 of 6

The brief and its four sections

Video V58

15 minutes

A brief settles the problem, the data, the measure and the risk — and deliberately leaves the algorithm out.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 24 pages of the video deck are in the professor's edition.

Pages from video V58

The One-Page AI Project Brief

Why 95% accuracy still fails projects — and the one document that prevents failure.



V58 page 1: The One-Page AI Project Brief

One Page.

Four Sections. Zero Ambiguity.

The structure that keeps AI projects on track — below a single line of code in Python.

V58 page 5: One Page. Four Sections. Zero Ambiguity.

The Four Required Sections

1	2
Task Type What kind of ML problem is this?	Data What exists today? What's missing?
3	4
Success Metric How will you know it's good enough?	Primary Risk What's the most likely way this fails?

V58 page 6: The Four Required Sections

A project brief is not paperwork. It is the written agreement about which problem is being solved, made before anyone builds anything.

What does “improve support efficiency” tell the team that has to build it?

Nothing they can build or test. One manager means faster replies, another means fewer staff, a third means fewer complaints. Work starts anyway, the model reaches 95 out of 100 correct in testing, and within weeks nobody in the business opens it.

the word: **deployment**

What does “sort each email into urgent, standard or low priority” tell them?

Three named categories. The past tickets can be prepared, the kind of model is settled, and the result can be measured next week. The disagreement happens now, on paper, where it costs an hour instead of a quarter.

the word: **classification**

Four sections settle it: the task type, the data, the success measure and the main risk. The algorithm is not one of them.



V58 page 10: Real-World Example: Customer Support Email Triage



V58 page 11: Section 1: Task Type

The Four Required Sections

1	2
Task Type What kind of ML problem is this?	Data What exists today? What's missing?
3	4
Success Metric How will you know it's good enough?	Primary Risk What's the most likely way this fails?

V58 page 6: The Four Required Sections

The memo that asked for shorter reports, London 1940

Step 1



August 1940, three months into the job

Winston Churchill is Prime Minister of the United Kingdom. Britain is under air attack, and almost every decision reaches the War Cabinet as a written paper.

Step 2



The problem he names

Ministers and their staff have to read a mass of papers, and nearly all of them are far too long. Time is lost, and energy goes into hunting for the points that actually matter.

Step 3



The memorandum, 9 August 1940

One page, titled Brevity. Set out the main points in short, crisp paragraphs. Move the detailed analysis to an appendix. Sometimes send the headings only, with the full report attached. Replace woolly phrases with plain words.

Step 4



What it costs, and what it buys

He accepts that the shorter style looks less formal. In exchange it saves time and, in his own description, forces clearer thinking. The discipline is on the writer, not the reader.

Step 5



What it leaves us

The document is still held at The National Archives. The one-page brief asks the same thing of an AI (artificial intelligence) project: state the four decisions, and put the workings behind them.

Three things to keep

1

A brief is judged by what it settles, not by how much it contains.

2

Four sections settle it: the task type, the data, the success measure and the main risk.

3

The algorithm is a later decision, taken by the technical team, and it is not part of the brief.

The Four Required Sections

1	2
Task Type What kind of ML problem is this?	Data What exists today? What's missing?
3	4
Success Metric How will you know it's good enough?	Primary Risk What's the most likely way this fails?

V58 page 6: The Four Required Sections

Important Notice

What the Brief Does NOT Ask

✗ Which algorithm?	✗ How many layers?
✗ What framework?	

The brief is a business clarity tool — not a technical specification.
Technical decisions come after the brief is solid.

V58 page 7: What the Brief Does NOT Ask



V58 page 8: Why One Page?

The head of customer support asks for an AI (artificial intelligence) tool to improve support efficiency. A supplier is ready to start on Monday, and the data team has been asked to export everything in the ticket system.

What has to happen before any of that work starts?

A

Choose the algorithm first, because that decision shapes everything else.

B

Write one sentence that says exactly what the model predicts, in named categories.

C

Export as much data as possible now and decide what it is for later.

D

Agree the budget and the delivery date, then let the supplier propose the rest.

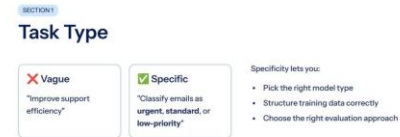
45 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.



V58 page 10: Real-World Example: Customer Support Email Triage



V58 page 11: Section 1: Task Type

Block 1, V58 Question 2 of 3

A supplier sends a two-page proposal. Page one names the model type and the number of layers. The data section says the team will collect what it needs as the project goes along. There is no number anywhere. The risk section says model performance may degrade.

Which part of this proposal is the most serious problem?

A

The model type is named far too early.

B

The data section describes what the team hopes to have, not what exists today.

C

The proposal runs to two pages instead of one.

D

Layer counts are too technical for the people who have to approve it.

45 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Pages from video V58

SECTION 2

Data

What exists today

3 years of labeled support tickets in the system

What's needed

Recent unlabeled tickets + response time logs from the past month

State what's available **right now** -- and where the gaps are. No wishful thinking.

V58 page 12: Section 2: Data

Is Your Brief Ready?

4 Red Flags to Check

→ Vague success criteria

"Improve accuracy" = not ready

→ Data section lists wishes

"We'll collect it somehow" = not a brief

→ Team can't agree on task or risk

Conflict now = expensive conflict later

→ Longer than one page

Scope is too broad or too fuzzy

V58 page 20: Is Your Brief Ready? Four Red Flags

Two versions of the success measure are on the table. The first says improve the accuracy of email sorting. The second says at least 92% precision on the emails flagged urgent. The support director asks why the second is worth an argument.

What does the second version give the team that the first does not?

A

It promises a better model than the first version.

B

It sets a test that can be run every week, and it says how many false alarms are acceptable: at most 8 in every 100 flagged.

C

It makes the project cheaper to build.

D

It tells the technical team which algorithm to use.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

SECTION 1

Success Metric

92% Precision

Of all emails flagged as urgent — at least 92% must genuinely be urgent.

Missing urgent emails violates customer service-level agreements. Numbers make this non-negotiable.

V58 page 13: Section 3: Success Metric

Success Metric → Business Accountability

✗ "improve accuracy"
Means nothing without context

✓ "92% precision on urgent"
Specific, measurable, business-grounded

A specific metric keeps the team pointed at what **genuinely matters** — not abstract statistical performance.

V58 page 18: Success Metric to Business Accountability

Your turn: cut the brief to one page

Eight things a team wanted in its brief, with the lines each takes. A page is 30 lines. Tick items on and off; the counts are written on the right.

What you can move on screen

Task type (3 lines) [on](#) or [off](#)

Section 1. What the model predicts.

Data that exists (4 lines) [on](#) or [off](#)

Section 2. What is in the systems now.

Success measure (3 lines) [on](#) or [off](#)

Section 3. Good enough, as a number.

Main risk (3 lines) [on](#) or [off](#)

Section 4. The likeliest way it fails.

Algorithm settings (6 lines) [on](#) or [off](#)

Decided later, by the technical team.

Project history (8 lines) [on](#) or [off](#)

Background only. It settles nothing.

Team and roles (5 lines) [on](#) or [off](#)

Belongs in the project plan, not here.

Glossary of terms (7 lines) [on](#) or [off](#)

Needed only if the wording is technical.

Try, in this order

1. Leave it as it opens. All eight items are in, and the brief runs past one page. Read the verdict and the number of lines.
2. Untick the glossary and the project history. Watch the page count fall, and check whether it fits yet.
3. Untick the team list and the algorithm section. Only the four decisions are left now; read how many lines they take and how much room is spare.

The point: The one-page limit is a filter. It removes everything that is not one of the four decisions, and the four decisions fit easily with room to spare.

The line counts are fixed working figures, chosen to behave like a real brief; they are not measurements of one document.

Three things from this block

1

Write the brief before the build. An hour of disagreement on paper is cheaper than a week of rebuilding.

2

Four sections, one page, no algorithm: task type, data, success measure, main risk.

3

If the four sections will not fit on one page, the problem is not defined yet.



V58 page 22: Clarity Saves Weeks



V58 page 23: The Complete Brief at a Glance



V58 page 24: One Page. Real Clarity.

Block 2 of 6

The brief filled in, and the red flags

Video V58

15 minutes

A brief is ready when a stranger could check every line against something that already exists.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 24 pages of the video deck are in the professor's edition.



V58 page 11: Section 1: Task Type



V58 page 13: Section 3: Success Metric



V58 page 20: Is Your Brief Ready? Four Red Flags

A brief is ready when someone outside the project could read it and verify every line without asking you a question.

What does a brief that is not ready look like?

Improve accuracy. We will pull the customer data we need as we go. The risk is that model performance may degrade. Nothing here can be checked, so nobody can disagree with it — and nobody can be held to it either.

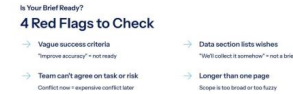
the word: **success metric**

What does a ready brief look like?

Sort each email into urgent, standard or low priority. Three years of answered tickets, plus last month's response records. At least 92% precision on urgent, so at most 8 false alarms in every 100 flagged. If urgent mail is missed, replies arrive late and customers leave.

the word: **precision**

Four red flags catch the rest: no number, data that is still a wish, no agreement on the task or the risk, and anything longer than one page.



V58 page 20: Is Your Brief Ready? Four Red Flags



V58 page 23: The Complete Brief at a Glance



V58 page 21: The Brief as a Shared Contract

The building that was started before it was designed, Sydney 1957 to 1973

Step 1



A competition, and a promise, 1957

The New South Wales government announces the winner of a design competition for an opera house in Sydney: Jørn Utzon, a Danish architect. In July it commits 7 million Australian dollars and a finish date of December 1963.

Step 2



Work starts, March 1959

Building begins on the base while the roof is still an idea on paper. The base later has to be altered to carry what was eventually designed.

Step 3



Twelve attempts at the roof

Between 1957 and 1963 the architect and the engineers work through twelve versions before they find shells that can actually be built, from repeated precast segments.

Step 4



The architect leaves, 1966

After years of dispute over money and control, Utzon resigns in February 1966 and leaves the country. Other people finish the interiors.

Step 5



What it cost

It opened on 20 October 1973, ten years late, at 102 million dollars against the 7 million agreed. The building is admired today, and the overrun was decided at the beginning, when the commitment was made before the unknowns were settled.

Three things to keep

1

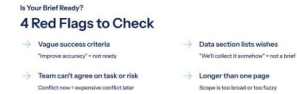
Ready means checkable: a reviewer can verify every line without asking you.

2

Four red flags: no number, data that is still a wish, no agreement on the task or the risk, more than one page.

3

Committing before the unknowns are settled does not remove the cost; it moves it to the end.



V58 page 20: Is Your Brief Ready? Four Red Flags



V58 page 21: The Brief as a Shared Contract



V58 page 22: Clarity Saves Weeks

A brief arrives for review. Task: reduce complaints using artificial intelligence. Data: we have several systems and will pull what we need. Success: fewer complaints. Risk: the model may not perform well.

Which line could a reviewer check against something that exists today?

A

The task line, because it names the technology that will be used.

B

The data line, because the systems it refers to do exist.

C

The success line, because complaints are already counted every month.

D

None of them. Every line states an intention rather than a fact or a threshold.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Is Your Brief Ready?

4 Red Flags to Check

- Vague success criteria
"Improve accuracy" = not ready
- Team can't agree on task or risk
Conflict now = expensive conflict later
- Data section lists wishes
"We'll collect it somehow" = not a brief
- Longer than one page
Scope is too broad or too fuzzy

V58 page 20: Is Your Brief Ready? Four Red Flags

SECTION 2

Data

What exists today

3 years of labeled support tickets in the system

What's needed

Recent unlabeled tickets + response time logs from the past month

State what's available **right now** — and where the gaps are. No wishful thinking.

V58 page 12: Section 2: Data

The data section of a brief reads: three years of answered support tickets in the ticket system; recent tickets from the past month with no answer recorded; response time records for the same month.

Why does this section separate what exists from what is still needed?

A

To make the brief longer and more convincing to the approver.

B

Because the missing part carries a cost and a timetable: somebody has to work through the unanswered tickets before training can start.

C

Because a model cannot use records older than one month.

D

Because the data is the technical team's decision, not the business team's.

40 seconds

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SECTION 2

Data

What exists today

3 years of labeled support tickets in the system

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State what's available **right now** — and where the gaps are. No wishful thinking.

V58 page 12: Section 2: Data

Data → Feasibility Check

Does the data exist?
Confirm before any modeling begins

Is it labeled?
Unlabeled data requires extra time and budget

Is it clean enough?
Garbage in, garbage out — always

V58 page 17: Data to Feasibility Check

Two managers read the same brief. One wants anything that might be urgent to be flagged. The other says every false alarm wastes an agent's time. The brief says: at least 92% precision on emails flagged urgent.

What does that line let them do that a discussion would not?

A

It proves the first manager is wrong.

B

It turns the disagreement into one number they can accept or change now: at most 8 false alarms in every 100 flagged.

C

It removes the need for the discussion altogether.

D

It hands the decision to the technical team.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

The Brief as a Shared Contract

Sets expectations

Everyone aligned before work starts

Grounds decisions

Reference point when disagreements arise

Saves weeks

Clarity now prevents rework later



V58 page 21: The Brief as a Shared Contract

SECTION 1

Success Metric

92% Precision

Of all emails flagged as urgent — at least 92% must genuinely be urgent.

☐ Missing urgent emails violates customer service-level agreements. Numbers make this non-negotiable.

V58 page 13: Section 3: Success Metric

Your turn: take the brief from vague to ready

The same four sections, each in three strengths. Set them and read how many red flags are left.

What you can move on screen

Section 1, task type Improve support efficiency, Use AI to handle emails better, Sort each email: urgent, standard, low

What the model predicts. Ready means named categories.

Section 2, data We have plenty of data, We will collect what we need, 3 years of answered tickets, plus last month

Ready means records that exist today, with a period.

Section 3, success measure Improve accuracy, As accurate as we can get, At least 92% precision on urgent

Ready means a number somebody can check weekly.

Section 4, main risk The model may not work, We will monitor performance, Missed urgent mail: late replies, customers leave

Ready means the business harm, named.

Try, in this order

1. Leave it as it opens, all four vague. Read how many red flags are up and how many checks a reviewer could run.
2. Set the success measure to the ready version. One red flag goes, and a number appears in the panel.
3. Now set the data line, then the task and the risk. Watch the verdict change only when all four are ready.

The point: A brief is not ready in parts. One wish left in the data line is enough to stop the hand-over.

Three things from this block

1

Ready means checkable: named categories, records that exist, a number, and the business harm.

2

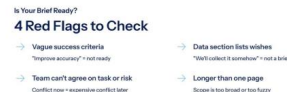
One vague section is enough to send the whole brief back.

3

Agreeing before the unknowns are settled moves the cost to the end, where it is largest.



V58 page 23: The Complete Brief at a Glance



V58 page 20: Is Your Brief Ready? Four Red Flags



V58 page 24: One Page. Real Clarity.

Block 3 of 6

Three traits of a problem worth solving

Video V59

15 minutes

A problem is worth solving when somebody acts on the output, the records exist, and the gain is worth more than the model.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 19 pages of the video deck are in the professor's edition.

Pages from video V59



V59 page 1: 93% Accurate. Completely Ignored.

3 Traits of a Solvable ML Problem



V59 page 5: Three Traits of a Solvable ML Problem

Do the Math



V59 page 8: Trait 3: Cost Structure, Do the Math

A model earns nothing until somebody does something differently because of it.

What happened at the bank?

The loan default model was right 93 times out of 100. The loan officers went on approving and rejecting exactly the same people, because nobody was required to use the score. The accuracy was real and the business result was zero.

the word: **accuracy**

What would have made it work?

A rule that the score changes something: a high score sends the file to a second reviewer, or the officer has to record a reason for going against it. The prediction has to arrive at a point where a decision is actually made.

the word: **actionable output**

Three traits: an output somebody acts on, enough history of reasonable quality, and a gain worth more than the model costs.

The Brutal Gap

Technical Success	if	Business Success
Model predicts well		
Business Success		
Changed decisions. Measurable value.		
Accuracy without adoption is worthless.		



V59 page 2: The Brutal Gap



TRAIT 1: ACTIONABLE OUTPUT

Prediction → Intervention

A model predicts which customers are at risk of canceling.
That output triggers a discount offer, a call, or targeted outreach.
No action = no value.

V59 page 6: Trait 1: Actionable Output

Do the Math

TRAIT 3: COST STRUCTURE		
2% Accuracy Gain Saves 5000/year	Model Cost \$500,000 to build & maintain	Verdict Don't build it. Value must exceed cost.

V59 page 8: Trait 3: Cost Structure, Do the Math

The telephone network that covered the whole planet, 1987 to 1999

Step 1



A plan drawn for the world of 1987

Motorola engineers design a ring of low-orbit satellites so that a telephone call can be made from anywhere on earth. At that time ordinary mobile networks cover very little ground.

Step 2



Sixty-six satellites, about five billion dollars

Building the system takes about eleven years. The engineering works: the coverage is genuinely worldwide, including oceans and deserts.

Step 3



Service opens, 1 November 1998

A handset costs around 3,000 dollars and a call costs several dollars a minute. The handsets are heavy and do not work indoors or inside a car.

Step 4



The sum nobody redid

While the satellites were being built, ordinary mobile networks spread across the places where the customers actually were. The company needed several hundred thousand subscribers. It had about 55,000.

Step 5



Nine months

On 13 August 1999 the company filed for bankruptcy, nine months after opening for business. The technology was not the problem. The value had been calculated for a world that no longer existed.

Three things to keep

1

If nobody is required to act on the output, the accuracy does not matter.

2

Enough history of reasonable quality: not perfect, but enough, and from a stable period.

3

Do the sum. The gain has to be worth more than the model costs to build and to run.

3 Traits of a Solvable ML Problem



V59 page 5: Three Traits of a Solvable ML Problem



V59 page 6: Trait 1: Actionable Output

Do the Math



V59 page 8: Trait 3: Cost Structure, Do the Math

A retailer builds a model that lists, every Monday, the customers most likely to stop buying. The list is emailed to a shared inbox. Nobody owns that inbox, and no campaign is attached to the list.

Which of the three traits fails here?

A

Trait two: there is not enough data.

B

Trait one: the output is not attached to any action.

C

The task is a goal rather than a prediction.

D

Trait three: the cost structure is negative.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

3 Traits of a Solvable ML Problem



V59 page 5: Three Traits of a Solvable ML Problem



V59 page 6: Trait 1: Actionable Output

A model would raise accuracy by two points. Each point is worth about 250 dollars a year in prevented losses. Building and running the model costs 200,000 dollars over three years.

What does the arithmetic say?

A

Build it. Two points of accuracy is a substantial gain.

B

Build it. The cost will come down once the model is running.

C

Do not build it. Three years of value is about 1,500 dollars against 200,000 dollars of cost.

D

There is not enough information to decide.

45 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Do the Math

TRAIT 3: COST STRUCTURE

2% Accuracy Gain

Saves \$500/year

Model Cost

\$200,000 to build & maintain

Verdict

Don't build it. Value must exceed cost.

V59 page 8: Trait 3: Cost Structure, Do the Math

3 Traits of a Solvable ML Problem

1

Actionable Output

The prediction directly informs a real decision someone acts on

2

Sufficient Data

Enough historical examples with reasonable quality to train on

3

Positive Cost Structure

Accuracy gains must translate to measurable business value

V59 page 5: Three Traits of a Solvable ML Problem

A subscription business has six months of records. Its prices and its customer mix changed twice in that period. A supplier says six months is plenty, because the table holds 40,000 rows.

What is wrong with the supplier's argument?



V59 page 7: Trait 2: Sufficient Data

3 Traits of a Solvable ML Problem



V59 page 5: Three Traits of a Solvable ML Problem

A

40,000 rows is far too few for any model.

B

The rows are many, but they cover a period in which the pattern itself changed twice, so the past does not describe the present.

C

Six months of customer data cannot be used legally.

D

Nothing is wrong. The number of rows is what matters.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Your turn: does the problem pass the three traits?

The bank's model with your numbers. Value and cost are both counted over three years.

What you can move on screen

What happens to the score Nobody is required to use it, It goes into a weekly report, High scores go to a second reviewer

Trait 1. Whether a decision changes because of the output.

History available 6 months of records, 18 months, 3 years, 5 years

Trait 2. Enough of a stable period to learn from.

Accuracy gain from 1 points to 10 points, in steps of 1

How much better than today's process.

Value of one point from \$250 a year to \$5000 a year, in steps of 250

What one point of accuracy saves in a year.

Cost to build and run \$50,000 over three years, \$120,000, \$200,000

Trait 3. Everything the model costs over three years.

Try, in this order

1. Leave it as it opens: the score goes nowhere, six months of records, two points worth 250 dollars each. Read the verdict.
2. Give the score an owner who must respond, and set the history to three years. Two traits now pass, and the money still does not.
3. Raise the value of a point until the three-year value passes the cost, and note how large it has to be.

The point: Two of the traits are about the organisation and one is arithmetic. All three have to pass, and the arithmetic is the one people skip.

The opening numbers are the video's: a two-point gain worth about 500 dollars a year, against a model costing 200,000 dollars.

Three things from this block

1

Trait one is organisational: somebody has to be required to act on the output.

2

Trait two is about the period covered, not only the number of rows.

3

Trait three is arithmetic, and it is the trait most often skipped.

3 Traits of a Solvable ML Problem



V59 page 5: Three Traits of a Solvable ML Problem

Do the Math



V59 page 8: Trait 3: Cost Structure, Do the Math

The Real Work Happens Here



V59 page 19: The Real Work Happens Here

Block 4 of 6

From a goal to a scoped task

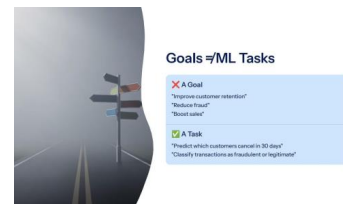
Video V59

15 minutes

A goal becomes a task when it names one prediction, one population, one time window and one decision.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 19 pages of the video deck are in the professor's edition.

Pages from video V59



V59 page 9: Goals Are Not ML Tasks

Where Teams Get Lost

- Predicting Too Far Ahead
30-day churn = tractable, 3-year churn = guessing
- Confusing Prediction with Decision
The model predicts churn; strategy decides what discount to offer.
- Underestimating Data Barriers
GDPR, HIPAA, retention policies — check access early.
- Overcomplicating Before Testing
Build the churn model first. Prove value. Then expand.

V59 page 12: The Four Pitfalls

The Well-Scoped Problem Statement

01 Prediction Task What exactly are you predicting?	02 Population Whose data trains the model? Who does it apply to?
03 Business Decision What action does the prediction inform?	04 Data Exists Historical records available? Legal access confirmed?

V59 page 18: The Well-Scoped Problem Statement

A model cannot solve a goal. It can answer one question, about one group of people, over one period.

Reduce customer churn.

Whose churn? Measured over what period? What happens when a name appears on the list? Three teams read that sentence and build three different things, and none of the three can be tested against it.

the word: **churn**

Predict which monthly subscribers will cancel within 30 days, so the retention team can offer them a call.

One prediction, one group, one window, one decision. The records can be assembled, the result can be checked next month, and somebody is waiting for the list.

the word: **prediction window**

Four elements make a scope: what is predicted, for whom, over what period, and which decision it feeds.



V59 page 9: Goals Are Not ML Tasks

Define Your Population



V59 page 11: Define Your Population

The Well-Scoped Problem Statement



V59 page 18: The Well-Scoped Problem Statement

The prize that turned a complaint into a test, London 1714

Step 1



A goal that had cost lives

For centuries a ship could work out how far north or south it was, but not how far east or west. Vessels ran aground, and cargoes and crews were lost.

Step 2



Parliament writes the task, 1714

The Longitude Act offers up to 20,000 pounds, a fortune at the time, for a method that fixes a ship's longitude to within half a degree, about 30 nautical miles, proved on a real voyage to the West Indies.

Step 3



A testable task attracts unexpected entrants

A board of commissioners judges entries against that threshold. Astronomers were expected to win. A self-taught clockmaker, John Harrison, entered with a sea clock instead.

Step 4



The voyage, November 1761

Harrison's fourth timepiece sailed to Jamaica with his son William. On arrival it was a few seconds out, which is far inside the limit the Act had set.

Step 5



What made it possible

The board argued about payment for another twelve years, but the test itself was never in dispute. A number, a method of checking and a named situation had turned an old complaint into a problem that could be solved.

Three things to keep

1

A goal names a direction. A task names one prediction that can be tested.

2

Name the population, because two groups with the same goal are two different problems.

3

A window that reaches too far turns a prediction into a guess.



V59 page 9: Goals Are Not ML Tasks

Define Your Population



V59 page 11: Define Your Population

Pitfall 1: The Time Window Trap

Stick to time windows where the future resembles the past.
Patterns that precede a 30-day cancellation are real signals. Two-year predictions are noise.



V59 page 13: Pitfall 1: The Time Window Trap

A board paper asks for a model to increase revenue from existing customers.

Which of these is a task a team could build against?

A

Increase revenue from existing customers by 5% this year.

B

Predict, for each monthly subscriber, whether they will buy the premium add-on within 60 days.

C

Apply machine learning across the customer base.

D

Understand our customers better.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.



V59 page 9: Goals Are Not ML Tasks

What Makes a Task Concrete?

- Narrow**
Focused on one specific prediction
- Measurable**
You can test whether it works
- Actionable**
Someone does something with the output

V59 page 10: What Makes a Task Concrete?

One cancellation model is planned for all customers: enterprise accounts on annual contracts and consumers paying month to month. All of them sit in the same table.

What is the strongest objection?

A

The table is too large to train on.

B

The two groups cancel for different reasons and on different timescales, so one model will fit neither of them well.

C

Enterprise accounts should never be modelled.

D

The model needs more columns.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Define Your Population



V59 page 11: Define Your Population

Your Secret Weapon: Domain Knowledge



V59 page 17: Your Secret Weapon: Domain Knowledge

A retention model produces a risk score for each customer. The project plan says the model will decide which customers receive a 20% discount, and when.

What is wrong with that plan?

A

Nothing. That is what automation means.

B

The model can produce the score, but who receives a discount, how large it is and when it is sent are business decisions with costs attached.

C

Discounts should never be offered to customers at risk of leaving.

D

The model would be too slow to decide in time.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Pitfall 2: Prediction $\neq$ Decision



A prediction is an input to a decision. Never the decision itself. Keep these roles separate.

V59 page 14: Pitfall 2: Prediction Is Not Decision

THE 4 PITFALLS

Where Teams Get Lost

- **Predicting Too Far Ahead**
30-day churn = tractable. 2-year churn = guessing.
- **Confusing Prediction with Decision**
The model predicts churn. Strategy decides what discount to offer.
- **Underestimating Data Barriers**
GDPR, HIPAA, retention policies — check access early.
- **Overcomplicating Before Testing**
Build the churn model first. Prove value. Then expand.

V59 page 12: The Four Pitfalls

Your turn: turn a goal into a scoped task

Four settings. The problem statement is written out underneath, with the pitfalls it triggers.

What you can move on screen

The goal you were given Improve customer retention, Reduce fraud, Boost sales
The business direction. It is never the task itself.

Population All customers together, Monthly subscribers, Enterprise accounts
Whose records train it and whose behaviour it predicts.

Time window The next 30 days, The next 6 months, The next 2 years
How far ahead the prediction reaches.

What the output feeds Nobody is required to act, A manager reviews the list and chooses offers,
The model sets the discount itself
The decision the prediction informs, and who owns it.

Try, in this order

1. As it opens, everything is at its loosest. Read the statement and the pitfalls it triggers.
2. Change the population to monthly subscribers, then shorten the window to 30 days. Two pitfalls go.
3. Give the list to a manager who must act on it, then read the finished statement aloud. That sentence is the first line of your brief.

The point: The statement is the scope. If you cannot say it in one sentence, the team will build several different things.

The pitfalls counted here are the four named in the video: too far ahead, prediction confused with decision, population not separated, and no action attached.

Three things from this block

1 One prediction, one population, one window, one decision.

2 Long windows and mixed populations are the two commonest ways a scope fails.

3 The model scores. A person decides what is done about the score.

The Well-Scoped Problem Statement

01	02
Prediction Task What exactly are you predicting?	Population Whose data trains the model? Whose does it apply to?
03	04
Business Decision What action does the prediction inform?	Data Exists Historical records available? Legal access confirmed?

V59 page 18: The Well-Scoped Problem Statement

Where Teams Get Lost

- **Predicting Too Far Ahead**
30-day churn = 1 tactic. 3-year churn = guessing.
- **Confusing Prediction with Decision**
The model predicts churn. Strategy decides what discount to offer.
- **Underestimating Data Barriers**
GDPR, HIPAA, retention policies — check access early.
- **Overcomplicating Before Testing**
Build the churn model first. Prove value. Then expand.

V59 page 12: The Four Pitfalls

The Real Work Happens Here

Validate your scope with domain experts, business leaders, and data engineers — before building anything.	
Right Problem Specific, actionable, data-backed	Right Team Domain + data + decision-model aligned
Real Impact Decisions change. Value is measurable.	



V59 page 19: The Real Work Happens Here

Block 5 of 6

Justifying the algorithm

Video V60

15 minutes

Name the constraint that decided the choice, and name what you rejected. That is a justification.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 25 pages of the video deck are in the professor's edition.

Weak vs. Strong Justification

Weak "Student housing performs best."	Strong "Student housing – interpretability isn't required and we have 500k labeled transactions."
---	---

Tie your choice to constraints, not just test scores.

V60 page 3: Weak versus Strong Justification

Three Dimensions Drive Every Choice

1. Interpretability Does regulation or policy require explainability?	2. Data Volume How many labeled examples do you have?
3. Task Type Classification, regression, clustering, or ranking?	



V60 page 4: Three Dimensions Drive Every Choice

Case B: Loan Approval

Interpretability Regulators require you to explain why a loan was denied
Consequence Black-box models become unjustifiable – regardless of accuracy
Interpretability is a hard constraint here. Accuracy alone cannot override it.



V60 page 10: Case B: Loan Approval

A choice becomes a justification the moment you name the constraint that made it, and say what you turned down.

Gradient boosting performs best.

The room hears a preference. Is it more than we need? Can it be explained to a customer? Was anything else tried? The sentence rests on a test score nobody present can check, so it invites doubt instead of settling it.

the word: **accuracy**

Gradient boosting, because no explanation is required here and we have 500,000 labelled transactions.

The same choice, now tied to two constraints anyone can verify. If either changes — a regulator arrives, the data turns out to be thinner — the room can see which way the decision moves.

the word: **interpretability**

Three dimensions decide it: must each result be explained, how many labelled examples exist, and what kind of task is this.

Weak vs. Strong Justification

Weak "Gradient boosting performs best." Their choice to maximize, not just test scores.	Strong "Gradient boosting — interpretability isn't required and we have 500k labelled transactions!"
--	--

V60 page 3: Weak versus Strong Justification

Dimension 1: Interpretability



V60 page 6: Dimension 1: Interpretability

Case B: Loan Approval

Interpretability Regulators require you to explain why a loan was denied
Consequence Black-box models become unjustifiable — regardless of accuracy
Interpretability is a hard constraint here. Accuracy alone cannot override it.



V60 page 10: Case B: Loan Approval

The reviewer who asked what the claim rested on, Washington 1960

Step 1



A routine file, September 1960

A new medical reviewer at the United States Food and Drug Administration, Frances Oldham Kelsey, is handed an application for a sedative already on sale in more than twenty countries. Everyone expects it to be signed within weeks.

Step 2



The questions she asks

She does not dispute the conclusion. She asks what it rests on: how the drug behaves in the body, and what evidence exists about its effect during pregnancy. A published report of nerve damage has not been explained.

Step 3



The pressure

The company answers, complains to her superiors and calls her an obstruction. She keeps asking for the missing evidence rather than for a different conclusion.

Step 4



The evidence arrives, late 1961

Serious harm to babies is reported in Europe and elsewhere. The application is withdrawn in March 1962, and the drug was never approved for general sale in the United States.

Step 5



What it changed

In August 1962 she received the President's Award for Distinguished Federal Civilian Service, and the law was tightened that year to require evidence before approval. Her standard was simple: not what do you conclude, but on what basis.

Three things to keep

1 A justification names the constraint, not the preference.

2 If a reason has to be given for each result, that requirement outranks the accuracy figure.

3 Saying what you rejected, and why, is what makes a choice defensible.

Weak vs. Strong Justification

Weak ✗ "Student housing performs best."	Strong ✓ "Student housing – interpretability isn't required and we have 500k labeled transactions."
---	---

Tie your choice to **constraints**, not just test scores.

V60 page 3: Weak versus Strong Justification

Dimension 1: Interpretability



Ask This First

- Does a regulator need to audit your model?
- Must you explain decisions to customers?
- Is there internal policy requiring transparency?

☐ If yes → interpretability is a **hard constraint**, not a preference.

V60 page 6: Dimension 1: Interpretability

The Framework in Summary



V60 page 24: The Framework in Summary

A lender compares two models for loan decisions. The complex one is two points more accurate. The regulator can require the lender to explain why any individual application was refused.

How should the written justification read?

A

The complex model, because two points of accuracy is a large gain.

B

The simpler model, because every refusal has to be explained, and that requirement is not traded against accuracy.

C

The complex model, with a note that explanations will be added later.

D

Whichever model the supplier recommends, since they know their product.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Dimension 1: Interpretability



Ask This First

- Does a regulator need to audit your model?
- Must you explain decisions to customers?
- Is there internal policy requiring transparency?

☐ If yes → interpretability is a hard constraint, not a preference.

V60 page 6: Dimension 1: Interpretability

Case B: Loan Approval

Interpretability

Regulators require you to explain why a loan was denied

Consequence

Black-box models become unjustifiable — regardless of accuracy

⚠ Interpretability is a hard constraint here. Accuracy alone cannot override it.



V60 page 10: Case B: Loan Approval

A factory wants to predict equipment failures. There are 2,000 labelled failure examples, and no regulator requires an explanation for individual predictions.

Which dimension is the limiting one?

A

Interpretability, because factories are regulated.

B

The amount of labelled data. 2,000 examples is thin for a complex model.

C

Task type, because failure prediction is not a category prediction.

D

None of them. All three point the same way.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Think before you proceed.

You're predicting equipment failure in a factory. You have 2,000 labelled examples and no regulatory requirement. Which dimension is your limiting factor?

Data Volume

2,000 examples — is that enough for complex models?

Interpretability

Not required — so this isn't the constraint

V60 page 11: Think Before You Proceed

Dimension 2: Data Volume & Quality

Data Volume Shapes Your Options

- Complex models need substantial labeled data
- Small datasets favor simpler, regularized methods
- Quality matters as much as quantity



V60 page 7: Dimension 2: Data Volume and Quality

Two sentences go to a steering group. First: our model achieved the highest score in testing. Second: we chose a tree-based model because the credit team must be able to read the rule behind each flag, and 12,000 rows is enough for it.

Why does the second sentence hold up better in the room?

A

It uses more technical vocabulary, which sounds authoritative.

B

It names the constraints that decided the choice, so anyone can check them and see what would change the answer.

C

It promises a better result than the first.

D

It avoids mentioning the data, which keeps the discussion simple.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Weak vs. Strong Justification

Weak

"Gradient boosting performs best."

Strong

"Gradient boosting — interpretability isn't required and we have 500k labeled transactions."

Tie your choice to **constraints**, not just test scores.

V60 page 3: Weak versus Strong Justification

Part 1 & 2: Algorithm + Metric

Algorithm Choice

- Name the algorithm
- Cite which dimension drove the decision
- Explain why you rejected alternatives

Metric Choice

- Name the metric
- State the cost asymmetry
- Connect to real business impact

A claim becomes an **argument** the moment you add a reason.

V60 page 17: Parts 1 and 2: Algorithm and Metric

Your turn: write the justification sentence

Three dimensions. The recommended method, the rejected alternative and the deciding constraint are written out for you.

What you can move on screen

Must each result be explained? Yes, a reason is required, No, the result is enough

Dimension 1. A regulator, a customer or a policy asking why.

Labelled examples 2,000, 10,000, 50,000, 500,000

Dimension 2. How many past cases carry an answer.

Task type Predict a category, Predict a number, Find groups, no answers recorded, Put items in order

Dimension 3. What kind of answer is wanted.

Try, in this order

1. As it opens, an explanation is required and there are 2,000 examples. Read which constraint decides it.
2. Set the explanation requirement to no. Watch the list of allowed methods grow and the deciding constraint move to the data.
3. Raise the data to 500,000 and read the finished sentence. That is the form your capstone justification takes.

The point: The deciding constraint is the tightest one, and it is the first thing your sentence should name.

The rules are the video's: a requirement to explain overrides accuracy, and complex methods need a large number of labelled examples before they are worth their cost.

Three things from this block

1

Name the tightest constraint first; it is what actually decided the choice.

2

A requirement to explain outranks a higher score.

3

State what you rejected. That is what turns a claim into an argument.

The Framework in Summary



V60 page 24: The Framework in Summary

Dimension 1: Interpretability



V60 page 6: Dimension 1: Interpretability

Part 3 & 4: Trade-offs + Plain Language



V60 page 18: Parts 3 and 4: Trade-offs and Plain Language

Block 6 of 6

Justifying the metric, and writing it down

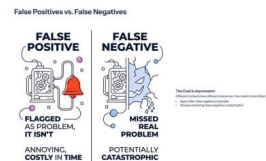
Video V60

15 minutes

The measure follows the money: whichever error costs more decides which number you optimise.

In this block: the one idea, a story, three things to keep, three questions, your turn, and three things from the block. All 25 pages of the video deck are in the professor's edition.

Pages from video V60



V60 page 13: False Positives versus False Negatives

Choosing the Right Metric

1	Accuracy Balanced classes, errors cost the same
2	Precision False positives are expensive – minimise false alarms
3	Recall False negatives are catastrophic – catch everything
4	F1 Score Errors matter, but neither dominates – balanced trade-off
5	AUC Ranking quality matters more than a fixed threshold

V60 page 14: Choosing the Right Metric

Metric Justification: F1 Score



V60 page 21: Metric Justification: F1 Score

The measure you choose is a decision about which mistake you are willing to make.

What does optimising accuracy do here?

Of 10,000 subscribers, 600 cancel. A model that flags nobody at all is right 94 times out of 100, and it loses the business all 600 customers. The number on the slide looks excellent and the loss is complete.

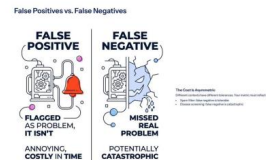
the word: **accuracy**

What do the two costs say?

A retention offer costs \$500 and a lost customer costs \$2,000: four to one. Real, but not extreme, so neither error dominates and a balanced measure fits. State the ratio and the measure follows from it.

the word: **F1 score**

Then write it down in four parts: the algorithm, the measure, the trade-offs you accepted, and a plain-language summary.



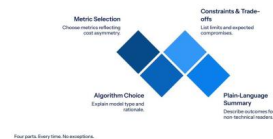
V60 page 13: False Positives versus False Negatives

Metric Justification: F1 Score



V60 page 21: Metric Justification: F1 Score

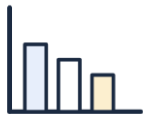
Structure Your Written Justification



V60 page 16: Part 3: Structure Your Written Justification

The taste test that answered the wrong question, Atlanta 1985

Step 1



A lead that had been slipping for years

By the early 1980s the company's share of its home market had been falling slowly for about fifteen years. Something had to be done, and the question was what.

Step 2



The measure they chose

Taste tests with nearly 200,000 people. The new, sweeter formula was preferred. By that measure the decision was settled before it was taken.

Step 3



23 April 1985

The original formula is withdrawn and the new one launched. The tests had answered exactly the question they were asked: which sample tastes better in a small cup.

Step 4



The question nobody asked

How people would feel about losing the original. Complaints, telephone calls and protest groups follow. The measure had no room for any of it, so none of it appeared in the results.

Step 5



Seventy-nine days

On 11 July 1985 the original returns as Coca-Cola Classic. The tests were not wrong. They measured a sip, and the business depended on something a sip does not capture.

Three things to keep

1 Ask which error costs more, then choose the measure.

2 Accuracy on an unbalanced problem can be high and worth nothing.

3 Write it in four parts: algorithm, measure, trade-offs, plain words.

Choosing the Right Metric

1	Accuracy Balanced classes, errors cost the same
2	Precision False positives are expensive – minimize false alarms
3	Recall False negatives are catastrophic – catch everything
4	F1 Score Balanced, but neither dominates – balanced trade-off
5	AUC Ranking quality matters more than a fixed threshold

V60 page 14: Choosing the Right Metric

Justify Your Metric With Costs

"We chose recall over precision because missing a fraud case costs 100x more than a false alarm."

You're not picking arbitrarily. You're matching the metric to business reality.



V60 page 15: Justify Your Metric With Costs

Structure Your Written Justification



V60 page 16: Part 3: Structure Your Written Justification

Of 10,000 monthly subscribers, 600 cancel each month. A model is built that flags nobody at all.

What is its accuracy, and what is it worth?

A 94% accurate, and worthless: it never identifies a single customer anyone could act on.

B 6% accurate, and worthless.

C 94% accurate, and good enough to start with.

D The accuracy cannot be worked out from these numbers.

45 seconds

Type the letter in the Zoom chat

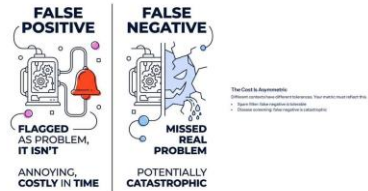
The answer is discussed in the session.

Choosing the Right Metric

1	Accuracy Balanced classes, errors cost the same
2	Precision False positives are expensive — minimize false alarms
3	Recall False negatives are catastrophic — catch everything
4	F1 Score Errors matter, but neither dominates — balanced trade-off
5	AUC Ranking quality matters more than a fixed threshold

V60 page 14: Choosing the Right Metric

False Positives vs. False Negatives



V60 page 13: False Positives versus False Negatives

A fraud team estimates that a missed fraudulent transaction costs about 100 times as much as a false alarm, which costs a few minutes of a reviewer's time.

Which measure should the justification name, and why?

A

Precision, because false alarms annoy the reviewers.

B

Recall, because the expensive error is the missed case, and recall is the share of real cases the model catches.

C

Accuracy, because it covers both kinds of error at once.

D

F1, because a balanced measure is always the safest choice.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.

Justify Your Metric With Costs

"We chose recall over precision because missing a fraud case costs 100x more than a false alarm."

You're not picking arbitrarily. You're matching the metric to business reality.



V60 page 15: Justify Your Metric With Costs

Choosing the Right Metric

1 Accuracy
Balanced classes, errors cost the same

2 Precision
False positives are expensive — minimize false alarms

3 Recall
False negatives are catastrophic — catch everything

4 F1 Score
Errors matter, but neither dominates — balanced trade-off

5 AUC
Ranking quality matters more than a fixed threshold

V60 page 14: Choosing the Right Metric

A capstone brief ends with this sentence: we used gradient boosting and reached an F1 score of 0.81.

What is missing from the justification?

A

Nothing. The number is the justification.

B

The constraints that decided the algorithm, the costs behind the measure, the trade-offs accepted, and a plain-language summary.

C

A fuller technical description of how gradient boosting works.

D

The names of the people who built it.

40 seconds

Type the letter in the Zoom chat

The answer is discussed in the session.



V60 page 16: Part 3: Structure Your Written Justification

Part 3 & 4: Trade-offs + Plain Language

Show What You Rejected

"We ruled out decision trees – interpretability alone doesn't justify their lower accuracy at this data volume."

Write for Non-Technical Readers

If only another data scientist understands it, it hasn't done its job. **Plain language = clear thinking.**

V60 page 18: Parts 3 and 4: Trade-offs and Plain Language

Your turn: let the costs choose the measure

10,000 subscribers a month, 600 of them cancel. Set the two costs and read what each setting costs the business.

What you can move on screen

Cost of one retention offer from \$100 to \$1500, in steps of 100

What it costs to contact one flagged customer.

Cost of one customer lost from \$500 to \$6000, in steps of 500

What the business loses when a cancellation is missed.

Measure you optimise Precision first, flag only the surest, Balanced, the F1 score, Recall first, flag widely, Accuracy first, flag nobody

The number the team is told to maximise.

Try, in this order

1. As it opens: \$500 an offer, \$2,000 a lost customer, a ratio of four to one. Read which setting is cheapest.
2. Raise the cost of a lost customer to \$6,000. The cheapest setting moves; note which way and why.
3. Now put the offer at \$1,500 and the loss back to \$2,000. The opposite setting wins. The measure was never a matter of taste.

The point: Nobody argues about the measure once the two costs are on the table. The ratio decides it.

How many customers each setting flags and catches are fixed working figures, chosen to behave like a real model. The 10,000 subscribers and 600 cancellations are the video's numbers.

Three things from this block

1

Two costs, one ratio: that is the whole argument about the measure.

2

Accuracy is the wrong measure whenever the outcome you care about is rare.

3

Write it in four parts, and finish with a plain-language summary.

The Framework in Summary



V60 page 24: The Framework in Summary

Metric Justification: F1 Score



V60 page 21: Metric Justification: F1 Score

Practice This Now

Apply the framework to your current or hypothetical project today.

Next: Communicating model limitations and uncertainty to non-technical audience - because every model has boundaries.



V60 page 25: Practice This Now

After today: the capstone

- Write your one-page brief with the four sections: task type, data, success measure with a number, main risk.
- Use the AI Solution Evaluation Matrix you submitted before Session 5 as the evidence behind it.
- Read two classmates' briefs against the four criteria and write at least two suggestions for each criterion.
- Keep the reviewers' comments; the final grade looks at what you changed and why.

From here: This was the last live session. Everything from here is written work: draft, two peer reviews, revision, final brief. Already behind you: the midterm exam on Modules 1 to 6, and Quiz 6 on Modules 11 and 12, which was due before today.

What is assessed, and when

What	Weight	When
Quiz 6 (Modules 11 and 12)	Part of 10%	Was due before today, 26 September 2026
Attendance and participation	10%	Recorded at each live session, including today
Capstone brief, first draft	Part of 50%	Thursday 1 October 2026
Two peer reviews written	Part of 50%	Sunday 4 October 2026
Revision after the feedback	Part of 50%	Wednesday 8 October 2026
Final brief submitted	50% in total	Saturday 10 October 2026, hard deadline